

CAMINA labelling guide

Nine road-user classes for a roadside counting sensor

UCD Spatial Dynamics Lab · Version 1.0, 25 September 2026

CAMINA counts the road users that cross a line in a camera's view. The detector learns exactly what the labels say, so one consistent rule matters more than many images. This guide fixes those rules: for correcting the current dataset and for labelling new images from Dublin cameras.

The current dataset (TRA 2026: 1,837 images, 13,148 boxes) was labelled by an automatic pipeline. A COCO detector supplied the common classes, an open-vocabulary detector (YOLO-World) supplied e-scooter, SUV and delivery_van, and cyclist and e-scooter boxes were built by joining a person to the vehicle beneath them. How many labels were then corrected by hand is not recorded. Every example below is from that dataset, shown with its labels as they are; Figure 4 shows errors that remain.

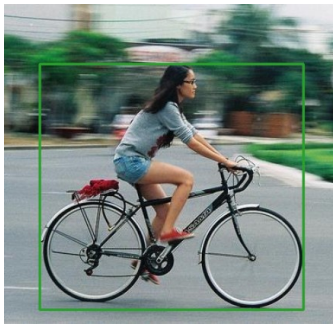
Five rules

1. **Label every road user in the image.** *Why:* an unlabelled road user teaches the model that it is background.
2. **One box per road user.** A rider and the bicycle, e-scooter or motorcycle they ride share one box; the rider gets no separate person box. *Why:* the sensor counts one track per road user, and a second box would count the rider twice. The dataset already does this for 97% of cyclists.
3. **Box what is visible, tightly.** Do not extend a box behind an occluding object. Label objects cut by the image edge. *Why:* the detector can only learn what it can see, and guessed extents differ from one annotator to the next.
4. **A rider class needs a rider.** A parked bicycle, e-scooter or motorcycle is not labelled; a person pushing one is a person. A rider stopped with a foot down is still a rider. *Why:* the classes describe how someone is travelling; a person walking a bicycle moves, and looks, like a pedestrian.
5. **When unsure, choose the broader class, or skip.** Between car and SUV, choose car. If you cannot tell what an object is, leave it unlabelled. *Why:* a wrong label on a rare class costs more than a missed one, and an object no one can identify cannot be labelled consistently.

The nine classes

Class	What it is	Not this	Boxes
■ person	Anyone on foot, standing or sitting; includes a person pushing a bicycle or scooter, on a skateboard, or in a wheelchair	People inside vehicles; people in posters or reflections	6,975
■ cyclist	A person riding any bicycle, e-bike or cargo bike; one box around rider and bicycle	A parked bicycle; a person walking a bicycle	2,012
■ car	A passenger car: saloon, hatchback, estate, coupé, MPV, taxi	SUVs and crossovers	2,105
■ e-scooter	A person standing on a kick scooter, electric or not; one box around rider and scooter	A parked or carried scooter; a seated moped rider	728
■ SUV	A sport utility vehicle or crossover: a tall body sitting high over the wheels, upright tailgate (Toyota RAV4, Nissan Qashqai, Hyundai Tucson)	MPVs (car); pickups (truck)	456
■ motorcyclist	A person riding a motorcycle or moped; one box around rider, passenger and vehicle	A parked motorcycle	307
■ bus	A bus or coach, single- or double-deck	A tram (Luas): not labelled	321
■ delivery_van	A van body, whatever it carries (Ford Transit, VW Transporter, Renault Kangoo)	A box body behind a separate cab (truck)	112
■ truck	A goods vehicle with a separate cab: box truck, lorry, tipper, bin lorry, pickup; an articulated lorry is one box	Vans	132

Three choices, and why. *Pickups are trucks*, as in COCO, so pre-labels from COCO-trained models need no correction there. *Trams are not labelled*: there is no tram class, and labelling the Luas as bus would add trams to the bus count. *Any van body is a delivery_van*: whether a van is delivering cannot be seen, and its shape can.



(a) cyclist: rider and bicycle in one box.



(b) e-scooter: rider and scooter.



(c) Stopped, one foot down: still e-scooter.



(d) motorcyclist: rider and moped.

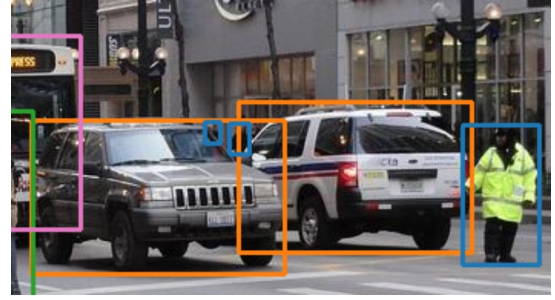
Figure 1: Rider classes: one box around the rider and the vehicle (rules 2 and 4).



(a) car: a low saloon body.



(b) SUV: a tall body, high over the wheels.

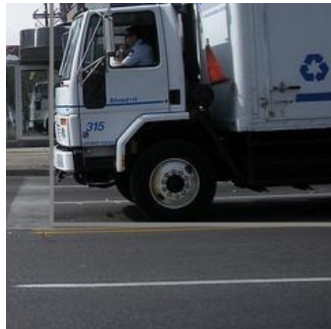


(c) Two SUVs; the officer is a person.

Figure 2: Car or SUV. Look at the height of the body over the wheels; if still unsure, choose car (rule 5).



(a) delivery_van: a van body.

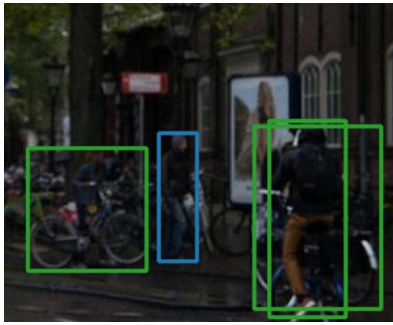


(b) truck: a box behind a separate cab.



(c) bus.

Figure 3: Vans, trucks and buses.



(a) Parked bicycles, and a person pushing one, labelled `cyclist`. Should be: nothing; person.



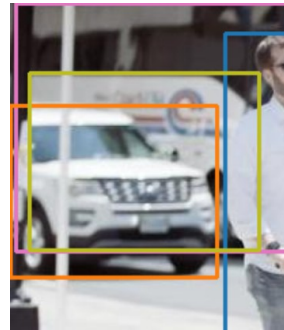
(b) An e-scooter rider labelled person. Should be: e-scooter.



(c) A car on its roof labelled truck. Should be: car.



(d) A tram labelled bus. Should be: nothing.



(e) One SUV with two boxes, SUV and `delivery_van`. Should be: one SUV box.

Figure 4: Errors that remain in the current dataset. Fix these when you meet them.

Workflow for new images

1. **Pre-label** with the most accurate model available; it runs on a GPU, so it need not be the Raspberry Pi model.
2. **Review every box:** correct its class and its extent, and delete boxes on things that are not road users.
3. **Look for what is missing**, especially small, distant and partly hidden road users. Pre-labels make it easy to accept what is there and forget what is not.
4. **Check one class at a time** over the whole batch (all SUVs, then all vans): inconsistencies stand out side by side.
5. **Keep the camera in the file name**, so the dataset split can keep each camera's images together and the test score stays honest.

The figures are generated from the committed labels by `docs/labelling_guide/render.py`; box counts are from `training/dataset/split.json`. The class list and its order are fixed by `configs/classes.yaml`.